

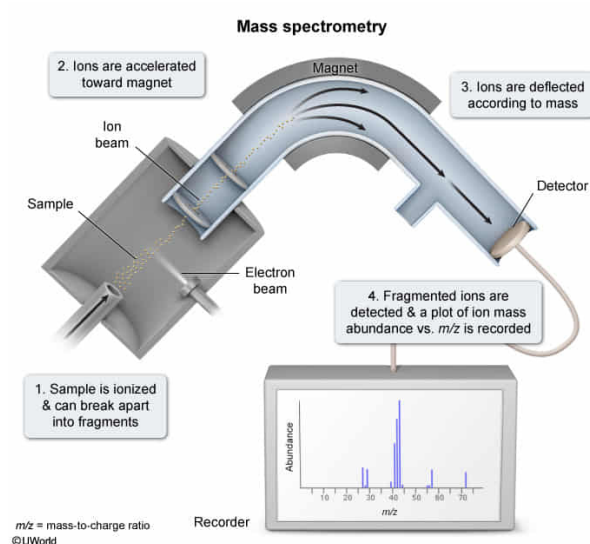
Which of the following statements correctly describes the peaks displayed in a mass spectrum?

- ☐ A. The tallest peak is the molecular ion. [9%]
- ☐ B. The peaks are uncharged molecules. [1%]
- ☐ C. The height of each peak is dependent on molecular weight. [26%]
- ✓ ☒ D. The peaks represent ionized fragments of the sample. [62%]

Correct

61% answered correctly

Explanation:



Mass spectrometry is a technique that measures the molecular weight of a molecule. Molecules in a sample are injected into a mass spectrometer, where they are **bombarded** with a beam of **electrons**. This beam removes electrons from the molecule and forms a positively charged **ion** known as the **molecular ion**. The molecular ion can also **fragment** during bombardment to form smaller ions.

An electric field accelerates the ions toward a magnet, which deflects them according to mass. The strength of the magnetic field is gradually changed during the experiment, and each field strength causes ions of a specific mass to reach the detector while all others are deflected into the walls of the tube. The ions are detected, and a mass spectrum is generated. The y-axis of the mass spectrum represents the **ion mass abundance**, and the x-axis represents the **mass-to-charge ratio** (m/z). The mass spectrum can be used to identify the mass of a molecule's fragments by taking the m/z difference between peaks.

Samples analyzed by mass spectrometry are ionized and fragmented before detection. Therefore, the peaks observed in the mass spectrum represent **ionized fragments** of the sample.

(Choice A) The peak that corresponds to the highest m/z ratio is typically the molecular ion. This peak *may*, or may not, be the tallest peak.

(Choice B) A sample analyzed by mass spectrometry is ionized when injected, and only ions are detected. Therefore, the peaks represent charged species rather than uncharged molecules.

(Choice C) Because the y -axis of a mass spectrum reflects abundance and the x -axis reflects molecular weight, the tallest peak represents the most abundant fragment. Therefore, peak height is *not dependent* on molecular weight but rather on abundance.

Educational objective:

Mass spectrometry is a technique that measures the molecular weight of a molecule. Molecules in a sample are bombarded with a beam of electrons, producing positively charged ions and fragments of the molecule. The ionized fragments are detected and a mass spectrum is generated, with the y -axis representing ion abundance and the x -axis representing the mass-to-charge ratio (m/z).

Time spent:0 sec

Subject:
Organic Chemistry

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System:
Separation Techniques, Spectroscopy, and Analytical
Methods